

Howard Math 273, Midterm Exam,

Fall 2026; Instructor: Sam Hopkins; Taken on: Wednesday, October 7th

Each problem is worth 10 points, for a total of 40 points. You have the full 80 minute class period to work on these problems. Partial credit will be given generously, so even if you cannot fully solve a problem, show me what you know.

1. Define the sequence of integers a_n by $a_0 = 1$, $a_1 = 2$, and $a_n = 2a_{n-1} + 3a_{n-2}$ for $n \geq 2$. So $a_n = 1, 2, 7, 20, 61, \dots$ for $n = 0, 1, 2, 3, 4, \dots$. Find the real numbers b and c for which we have $a_n \sim c \cdot b^n$ as $n \rightarrow \infty$.
2. Define the sequence of integers c_n by $c_0 = 1$ and $c_n = \sum_{k=0}^{n-1} c_k \cdot c_{n-1-k}$ for $n \geq 1$. So $c_n = 1, 1, 2, 5, 14, \dots$ for $n = 0, 1, 2, 3, 4, \dots$. (These are called the *Catalan numbers*.)
 - (a) Form the generating function $\mathcal{C}(x) := \sum_{n \geq 0} c_n x^n$. Argue, using the defining recurrence for the c_n , that $\mathcal{C}(x) = 1 + x \cdot \mathcal{C}(x)^2$.
 - (b) Use the quadratic formula to conclude that $\mathcal{C}(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$. (We can show the $\pm$ must be a $-$ by checking the first term in the series expansion, but you don't have to do this.)
 - (c) Show that $1 - \sqrt{1-4x} = \sum_{n \geq 0} \frac{2}{n+1} \binom{2n}{n} x^{n+1}$. **Hint:** You can use, without proof, that $\frac{1}{\sqrt{1-4x}} = \sum_{n \geq 0} \binom{2n}{n} x^n$ (which we showed in class), together with a derivative/integral.
 - (d) Conclude that $c_n = \frac{1}{n+1} \binom{2n}{n}$ for all $n \geq 0$.
3.
 - (a) Let a_n be the number of partitions of n where, for each $k \geq 1$, there are at most two parts equal to k . Show that $\sum_{n \geq 0} a_n x^n = \prod_{k \geq 1} (1 + x^k + x^{2k})$.
 - (b) Let b_n be the number of partitions of n where each part is congruent to 1 or 2 modulo 3. Show that $\sum_{n \geq 0} b_n x^n = \prod_{k \geq 1} \frac{1}{1-x^{3k-2}} \cdot \prod_{k \geq 1} \frac{1}{1-x^{3k-1}}$.
 - (c) Conclude that $a_n = b_n$ for all $n \geq 0$. **Hint:** Use parts (a) and (b), and the algebraic identity $(1-x^3) = (1+x+x^2)(1-x)$.
4. Show that, for each $n \geq 2$, the number of permutations $\sigma \in S_n$ for which 1 and 2 belong to the same cycle of σ is $\frac{n!}{2}$. **Hint:** Use induction. Check the base case of $n = 2$. For the induction step, imagine building a permutation in S_n from one in S_{n-1} by writing the permutations in cycle notation and placing the n somewhere.